

CURRENT ELECTRICITY

Quick Reference – Theory & Formulas

1. ELECTRIC CURRENT & CHARGE FLOW

Electric Current (I): Rate of flow of electric charge

- Formula: $I = \frac{Q}{t}$
- Unit: Ampere (A)
- $1 \text{ A} = 1 \text{ C/s}$

Charge (Q):

- $Q = ne$, where n = number of electrons, $e = 1.6 \times 10^{-19} \text{ C}$
- Unit: Coulomb (C)

Current Direction: Opposite to electron flow (conventional current)

Types of Current:

- **DC:** Constant magnitude and direction
 - **AC:** Alternating direction, time-varying
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2. DRIFT VELOCITY & MOBILITY

Drift Velocity (v_d)

- $v_d = \frac{I}{nAe}$
- n = number density of electrons
- A = cross-sectional area
- e = electron charge

Mobility (μ)

- $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$
- τ = relaxation time
- m = electron mass

Relation: $I = nAe\mu E$

3. RESISTANCE & RESISTIVITY

Resistance (R):

- $R = \frac{V}{I}$
- Unit: Ohm (Ω)

Resistivity (ρ):

- $R = \rho \frac{L}{A}$
- Unit: $\Omega \cdot m$

Conductivity (σ):

- $\sigma = \frac{1}{\rho}$

Variation with Temperature:

- $R_T = R_0[1 + \alpha(T - T_0)]$
 - $\rho_T = \rho_0[1 + \alpha(T - T_0)]$
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4. EMF & INTERNAL RESISTANCE

EMF ($\mathcal{E}$):

- $\mathcal{E} = \frac{W}{Q}$
- Unit: Volt (V)

Terminal Voltage:

- Discharging: $V = \mathcal{E} - Ir$
- Charging: $V = \mathcal{E} + Ir$

Current from Battery:

- $I = \frac{\mathcal{E}}{R+r}$
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5. OHM'S LAW & KIRCHHOFF'S LAWS

Ohm's Law: $V = IR$

Microscopic form: $\vec{J} = \sigma \vec{E}$

Kirchhoff's Current Law (KCL):

- $\sum I_{in} = \sum I_{out}$

Kirchhoff's Voltage Law (KVL):

- $\sum \mathcal{E} = \sum IR$
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6. SERIES & PARALLEL RESISTANCES

Series:

- $R_{eq} = R_1 + R_2 + R_3 + \dots$

Parallel:

- $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

Current Divider:

- $I_1 = I \frac{R_2}{R_1 + R_2}$

Voltage Divider:

- $V_1 = V \frac{R_1}{R_1 + R_2}$

7. ELECTRICAL ENERGY & POWER

Power:

- $P = VI = I^2R = \frac{V^2}{R}$

Energy:

- $W = Pt = VIt$

Heat (Joule Heating):

- $H = I^2Rt$

Source Power:

- $P_{\text{source}} = \varepsilon I$

External Power:

- $P_{\text{ext}} = VI = I^2R$

Internal Power:

- $P_{\text{internal}} = I^2r$

Efficiency:

- $\eta = \frac{R}{R+r}$

8. COMBINATION OF CELLS

Cells in Series:

- $\varepsilon_{\text{total}} = \varepsilon_1 + \varepsilon_2 + \dots$
- $r_{\text{total}} = r_1 + r_2 + \dots$
- $I = \frac{\varepsilon_{\text{total}}}{R + r_{\text{total}}}$

Cells in Parallel (identical):

- $\varepsilon = \varepsilon_1 = \varepsilon_2 = \dots$
- $\frac{1}{r_{\text{total}}} = \frac{1}{r_1} + \frac{1}{r_2} + \dots$

Mixed Combination:

- EMF: $\varepsilon = m\varepsilon_1$
- Internal resistance: $r = \frac{mr_1}{n}$

9. POTENTIOMETER & METER BRIDGE

Potentiometer Principle:

- $\frac{E_1}{E_2} = \frac{L_1}{L_2}$

Uses:

- Measure EMF

- Compare EMFs
- Find internal resistance

Meter Bridge:

- At balance: $\frac{R_1}{R_2} = \frac{R_3}{R_4}$
 - $\frac{R_1}{X} = \frac{l_1}{l_2}$
 - $l_1 + l_2 = 100 \text{ cm}$
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10. AMMETER & VOLTMETER

Ammeter:

- Low resistance
- Series connection
- Ideal ammeter: $R_A = 0$

Voltmeter:

- High resistance
- Parallel connection
- Ideal voltmeter: $R_V \rightarrow \infty$

Shunt (Ammeter):

- $S = \frac{R_g}{n-1}$

Multiplier (Voltmeter):

- $M = R_g(n - 1)$
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11. SUPERCONDUCTIVITY

Superconductor: Zero resistance below critical temperature T_c

Properties:

- $\rho = 0$
- Perfect conductor
- Meissner effect

Critical Current: I_c

12. SEMICONDUCTORS & DIODES

Semiconductors: (Si, Ge, GaAs)

Resistivity: 10^{-3} to $10^5 \Omega \cdot m$

Intrinsic:

- $n_e = n_h = n_i$

Extrinsic:

- **N-type:** Donor (Group V), electrons majority
- **P-type:** Acceptor (Group III), holes majority

PN Junction:

- Forward bias: Conducts
 - Reverse bias: Blocks
 - Cut-in voltage: 0.6–0.7 V (Si), 0.3 V (Ge)
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13. THERMOELECTRIC EFFECTS

Seebeck Effect:

- $\varepsilon = S\Delta T$

Peltier Effect:

- $Q = \pi I$

Thomson Effect:

- Heating/cooling along conductor with temperature gradient
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14. ELECTRONIC CONDUCTION

Classical Model:

- Electrons move randomly; drift velocity is very small

Quantum Model:

- Band theory, conduction & valence bands

Fermi Level:

- Highest occupied state at 0 K
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15. IMPORTANT CONSTANTS & VALUES

Quantity	Symbol	Value
Electron charge	e	$1.6 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Avogadro number	N_A	$6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$

QUICK SUMMARY TABLE

Concept	Formula	Unit
Current	$I = Q/t$	A
Resistance	$R = V/I$	Ω
Resistivity	$\rho = RA/L$	$\Omega\cdot\text{m}$
Power	$P = VI = I^2R$	W
Energy	$W = Pt$	J
EMF	$\varepsilon = IR + Ir$	V
Drift velocity	$v_d = I/(nAe)$	m/s
Mobility	$\mu = v_d/E$	$\text{m}^2/(\text{V}\cdot\text{s})$